

Logistic Regression

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Course: Deep Learning and Neural Network

Course Code: CS4152

November 6, 2025

What is Logistic Regression?

- Supervised learning algorithm for **binary classification**.
- Labels: $Y \in \{0, 1\}$ (e.g., Cat = 1, Not-Cat = 0).
- Predict the probability of class 1 given features X :

$$\hat{y} = P(Y = 1 | X).$$

Model Definition

Linear score

$$z = W^T X + b, \quad W \in \mathbb{R}^{n_x}, \quad b \in \mathbb{R}.$$

Sigmoid link

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}} \in (0, 1).$$

- $\sigma(\cdot)$ maps any real number to a probability.
- Keeps outputs bounded and differentiable.

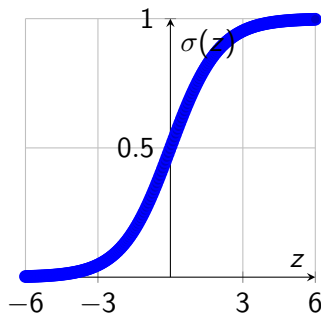
Why Not Linear Regression for Classification?

- Linear regression: $\tilde{y} = W^T X + b$ can output values < 0 or > 1 .
- Not a valid probability model; sensitive to outliers.
- Logistic regression fixes this with the **sigmoid** transformation.

Sigmoid Function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- $\sigma(0) = 0.5$.
- $z \rightarrow +\infty \Rightarrow \sigma(z) \rightarrow 1$.
- $z \rightarrow -\infty \Rightarrow \sigma(z) \rightarrow 0$.



Interpreting $z = W^T X + b$

- z is a **linear score**; $\sigma(z)$ maps score $\rightarrow$ probability.
- Large $z \Rightarrow$ high confidence for class 1 (cat).
- Very negative $z \Rightarrow$ high confidence for class 0 (not-cat).
- Decision boundary at $\hat{y} = 0.5$ (equivalently $z = 0$).

Parameters and Learning Target

- Parameters: $W \in \mathbb{R}^{n_x}$, $b \in \mathbb{R}$.
- Objective: choose (W, b) so that $\hat{y}$ approximates $P(Y = 1 | X)$.
- Leads to defining a **cost function** and using **gradient-based optimization**.

Notation Clarification

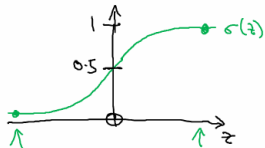
- In this course we keep W and b as **separate** parameters.
- Alternative (not used here): augment features with $X_0 = 1$ and write $\hat{y} = \sigma(\theta^T X)$.
- Keeping W and b separate simplifies implementation and vectorization.

Logistic Regression

Given x , want $\hat{y} = \frac{P(y=1|x)}{0 \leq \hat{y} \leq 1}$
 $x \in \mathbb{R}^{n_x}$

Parameters: $\underline{w} \in \mathbb{R}^{n_x}$, $\underline{b} \in \mathbb{R}$.

Output $\hat{y} = \sigma(\underbrace{w^T x + b}_z)$



$$x_0 = 1, \quad x \in \mathbb{R}^{n_x+1}$$
$$\hat{y} = \sigma(\theta^T x)$$

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \vdots \\ \theta_{n_x} \end{bmatrix} \begin{matrix} \} b \leftarrow \\ \} w \leftarrow \end{matrix}$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\text{If } z \text{ large } \sigma(z) \approx \frac{1}{1+0} = 1$$

If z large negative number

$$\sigma(z) = \frac{1}{1 + e^{-z}} \approx \frac{1}{1 + \text{Big num}} \approx 0$$

Andrew Ng

Summary and Next Steps

- Logistic regression models $P(Y = 1 \mid X)$ using $\sigma(W^T X + b)$.
- Sigmoid ensures probabilities in $(0, 1)$.
- Next: define the **cost function** and derive **gradients** for learning.